Dr. Sapna Mishra

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RESEARCH INTERESTS

• Satellite–host galaxy interactions. • Magellanic Clouds and Stream. • Galaxy evolution in the Local Group. • Circumgalactic and Intergalactic Medium. • Outskirts of the Galaxy Cluster. • AGN feedback. • Quasar absorption line spectroscopy.

ACADEMIC POSITIONS

- **Postdoctoral Fellow, 2023–present**, Milky Way Halo Research Group, Space Telescope Science Institute (STScI), Baltimore, USA.
- **Postdoctoral Fellow, 2021–2023**, Inter-University Centre for Astronomy & Astrophysics (IUCAA), Pune, India.
- **Post-Thesis Submission Fellow, 2020–2021**, **A**ryabhatta **R**esearch **I**nstitute of observational scienc**ES** (ARIES), Nainital, India.

EDUCATION

- **Ph.D., Astronomy, 2015–2020**, ARIES, Nainital, India. *Thesis: Probing the Environment of AGN Based on Their Feedback Processes*; Advisor: Prof. Hum Chand.
- **Pre-Ph.D. Coursework in Astronomy, 2014–2015**, ARIES, Nainital, India.
- **Master of Science (MSc), Physics, 2012–2014**, Department of Physics & Astrophysics, Delhi University (DU), India.
- **Bachelor of Science (BSc), Physics Honors, 2009–2012**, Miranda House College, DU, India.

RESEARCH GRANTS

- **2026:** JWST Cycle 5 (PI: T. Fischer), GO–10815.
- **2025:** HST Cycle 33 (PI: S. Mishra), GO+AR–18076, *Funding* $\sim \$99K$.
- **2025:** HST Cycle 33 (PI: S. Mishra), GO+AR–18072, *Funding* $\sim \$84K$.
- **2025:** HST Cycle 33 (PI: A. Fox), AR–18149, *Funding* $\sim \$29K$.
- **2024:** HST Cycle 32 (PI: S. Mishra), GO–17757, *Funding* $\sim \$110K$.
- **2024:** Simons Foundation — Support to attend the CGM Workshop, Aspen Center for Physics; *Funding* $\sim$ \$3,000.
- **2023:** FONDECYT Postdoctoral Fellowship 2023 (Chile), *Program #3230509, Funding $110,400.*
- **2022:** Milano–Bicocca Research Grant (Type A2) (Italy).

TELESCOPE TIME AS PRINCIPAL INVESTIGATOR

- **VLT/UVES (Cycle P117): 35 hours**, *The Distance to the Magellanic Stream Using RR Lyrae Stars, PID: 117.2A4J.*
- **HST/COS (Cycle 33): 30 orbits**, *Unveiling the Circumgalactic Medium of M33, PID: GO+AR-18076.*
- **HST/COS (Cycle 33): 22 orbits**, *The Fate of the Leading Arm of the Magellanic Stream, PID: GO+AR-18072.*
- **HST/COS (Cycle 32): 29 orbits**, *Probing the Front-Side of the Circumgalactic Medium of the Large Magellanic Cloud, PID: GO-17757.*
- **VLT/FORS2 (Cycle P109): 17 hours**, *Mg* II *Tomography of Cluster Outskirts Using 11 Background Quasars, PID: 109.23G6.*
- **Devasthal Optical Telescope (DOT, 3.6m, Cycle 22A): 2 Nights**, *NIR Spectroscopy of Post-starburst Galaxies to Probe Obscured Star Formation and Stellar Populations, PID: DOT-2022-C1-P18.*
- **DOT (Cycle 21A): 4 Nights**, *Probing the Connection Between Emission and Absorption Outflows in IR-bright BAL Quasars, PID: DOT-2021-C1-P32.*

- **DOT (Cycle 2018A): 16 Hours**, *Resolving the Narrow Emission-Line Region of the Quadruply Imaged Quasar RXS J113155.4−123155, PID: P325-2018A.*
- **DOT (Cycle 2017A): 4 Nights**, *Infrared Properties of Jet-dominated BALQSOs, PID: P31-2017A.*
- **Himalayan Chandra Telescope (HCT, 2m, Cycle 21-C2): 3 Nights**, *Probing Spectral Variability of X-ray Bright High-ionization BAL Quasars, PID: HCT-2021-C2-P56.*
- **HCT (Cycle 21-C1): 3.5 Nights**, *Intranight Monitoring of Blazar Counterparts of BAL Quasars, PID: HCT-2021-C1-P52.*
- **HCT (Multiple cycles): ≈12 Nights**, *Probing Environments of Emerging Broad Absorption Line Quasars, PIDs: HCT-2020-C2-P27; HCT-2020-C1-P170; HCT-2019-C3-P117.*

AS CO-INVESTIGATOR

- **JWST (Cycle 5): 14.6 hours**, *Following the Shocks: JWST IFU Spectroscopy of NGC 4258's Radio Arcs*; *PID: GO–10815.*
- **NRAO/GBT (Cycle 26A): 370 hours**, *Mapping Cool Clouds in the Milky Way's Nuclear Wind*; *PID: GBT26A-224.*

PRESS COVERAGE

- Mishra et al. (2024b) featured as a question on the **American TV quiz show *Jeopardy!*** (Feb 12, 2025).
- NASA's Hubble Sees Aftermath of Galaxy's Scrape with Milky Way — NASA press release.
- Hubble sees aftermath: Encounter blew away most of smaller galaxy's gaseous halo — ESA/Hubble press release.

SKILLS

- **Observing Experience:** ∼100 nights with 1–4m ground-based telescopes.
- **Techniques:** Spectroscopy (expert, 8 first-author papers), Photometry (proficient, 2 first-author papers), Astrometry (calibration-level experience, 1 proceeding).
- **Spectrographs Used:** HST/COS; VLT (UVES, X-shooter, FORS2); Keck/LRIS; SDSS/BOSS; SAO/SCORPIO; HST/HFOSC, DOT/DFOT.
- **Programming Languages:** `Python`, `IDL`, C, C++, `ecl-IRAF` scripts, Unix shell scripts.
- **Web Programming:** `PHP`, `MySQL`, `HTML`.
- **Astronomy Software:** `IRAF`, `DAOPHOT`, `CLOUDY`, `TOPCAT`, `ESOReflex/EsoRex`, `Gasgano`, `vpfit`.
- **Large-Dataset Queries:** SQL CasJobs queries for `SDSS`, `HST-MAST`, `SIMBAD`, `NED`.

PROFESSIONAL & ACADEMIC SERVICE

- **Panel Support Scientist**, HST/TAC, Cycle 34, STScI (2026).
- **LOC**, Spring Symposium, STScI (2026).
- **Panelist**, National Science Foundation (NSF), Remote (2026).
- **Reviewer & Mentor**, Space Astronomy Summer Program (SASP), STScI (2026).
- **Member**, American Astronomical Society.
- **Referee**, ApJ, ApJL.
- **Member**, Working Group on the High-Resolution Multi-Object Spectrograph (HRMOS) for the VLT.
- **Reviewer**, Chambliss Student Poster Competition, 247th AAS Meeting (2026).
- **Reviewer**, Space Astronomy Summer Program (SASP), STScI (2025).
- **Panel Support Scientist**, HST/TAC, Cycle 33, STScI (2025).
- **Organizing Committee Member**, STScI HotSci Colloquium Series (2024).
- **Panel Support Scientist**, HST/TAC, Cycle 32, STScI (2024).
- **Panel Support Scientist**, JWST Telescope Allocation Committee (TAC), Cycle 3, STScI (2024).
- **Service Observer**, Devasthal Optical Telescope (3.6m) — conducted observations on behalf of external proposers during the COVID-19 period.

MENTORING

- **June 2026–present: Noela Weramundi**, Space Astronomy Summer Program (SASP-2026).
- **Feb 2025–present: Zhibin You**, Undergraduate student, Johns Hopkins University.
  Research project: *"Probing the Small-Scale Routly–Spitzer Effect in the Milky Way ISM."*
- **Aug 2025–present: Khushi Mehta** (remote), Central University of Himachal Pradesh (India).
  Master's dissertation: *"Probing the Circumgalactic Medium of Virgo Cluster Galaxies."*
- **2023–2025: Ritish Kumar** (remote), Central University of Himachal Pradesh (India).
  Graduate research project: *"On the Incidence of Weak and Strong Mg II Absorbers Toward Flat and Steep Spectrum Radio Quasars."* Published in MNRAS (2025).
- **2017–2018: Sri Devi** and **Parth Nair**, Master's students in Physics.
  Summer dissertation projects: *"Photometric and Spectroscopic Variability of BAL Quasars."*

TEACHING & OUTREACH

- **Graduate-Level Instructor, IUCAA, India** — Taught three online graduate-course lectures on Astronomical Instruments (June 2022).
- **Lecturer, ARIES Training School in Observational Astronomy (ATSOA)** — Delivered spectroscopy lectures (theory and observation) in 2016, 2017, 2018, and 2019.
- **Hands-on Data-Reduction Instructor, ATSOA** — Provided 14-day intensive training in optical data reduction (2016–2019).
- **Instructor, TMT Workshop on Large Telescope Data Handling (IUCAA)** — Conducted training in high-resolution UVES spectral data reduction (Jan 15–27, 2017).
- **Mentor for Incoming Ph.D. Students, ARIES, India** — Guided students in telescope operations, observing techniques, and data handling.

Outreach & Public Engagement

- Astronomy for Non-Astronomers, STScI, June 21, 2026 (invited public talk).
- Delivered invited **public talks** at "Astronomy on Tap" (Baltimore, 2025) and undergraduate outreach lectures.
- Led **public sky-gazing sessions** using 1–2m class telescopes at ARIES and Devasthal observatories, India.

AWARDS & HONORS

- **2023:** FONDECYT-2023 Postdoctoral Prize Fellowship, Chile.
- **2022:** Milano-Bicocca Research Grant (Type A2), Italy.
- **2014:** All-India "Graduate Aptitude Test in Engineering" (GATE), India.
- **2012:** All-India "Joint Admission Test for Masters" (JAM), India.
- **2012:** Selected in the top 10% of graduate-level students, Delhi University.

SCIENTIFIC PRESENTATIONS

- **2026:**
  - FOGGIE Retreat Meeting, Mt. Washington, Baltimore, June 1, 2026.
  - Astronomy for Non-Astronomers, STScI, June 21, 2026 (invited public talk).
  - Galaxies and AGN Seminar and Journal Club, JHU & STScI, Feb 24, 2026 (invited, declined).
  - Centre for Particle Physics and Astronomy, Royal Holloway, University of London, Jan 28, 2026 (invited, declined).
  - 247th American Astronomical Society Meeting, Phoenix, Arizona, Jan 2026 (talk).
  - 247th American Astronomical Society Meeting, Phoenix, Arizona, Jan 2026 (splinter session).
- **2025:**
  - Aix-Marseille Université, "Workshop on the Network for Ultraviolet Astronomy," Oct 2025.
  - Space Telescope Science Institute, "HotSci Colloquium Series," July 2025.
  - Montana State University, "XMCII: Milky Clouds over Yellowstone," May 2025.
  - Space Telescope Science Institute, "Spring Symposium," May 2025 (poster/flash talk).
  - Summer Conference 2025, Ninth Edition, May 2025 (remote, invited talk).

- **2024:**
  - Aspen Center for Physics, “Holistic Picture of the CGM,” September 2024.
  - Center for Astrophysics | Harvard & Smithsonian, “Multiphase Madness,” August 2024.
  - Space Telescope Science Institute, “Spring Symposium,” April 2024.
  - Flatiron Institute, “XMCI: Milky Clouds over Manhattan,” February 2024.
  - Space Telescope Science Institute, “Galaxy/AGN Journal Club,” January 2024.
  - Space Telescope Science Institute, “CoolSci,” January 2024.
- **2023:** IUCAA, India, “Galactic Inflows and Outflows on All Scales,” February 2023.
- **2022:**
  - Università Milano-Bicocca, Milan, “What Matter(s) Around Galaxies,” September 2022.
  - IUCAA, India, “Monthly Last Friday Talk Series,” January 2022.
- **Pre-2020:**
  - IISER Tirupati, India, “Astronomical Society of India Meeting,” March 2020 (poster).
  - Department of Physics & Astrophysics, Delhi University, “Departmental Talk,” October 2019 (invited).
  - IUCAA, India, “Recent Trends in the Study of Compact Objects: Theory and Observations (RETCO-IV),” April 2019 (poster).
  - ARIES, India, “ARIES Training School in Observational Astronomy (ATSOA),” March 2019.
  - Institut d’Astrophysique de Paris (IAP), Paris, France, “Massive Black Holes in Evolving Galaxies: From Quasars to Quiescence,” May 2018 (poster/flash talk).
  - ARIES, India, “ARIES Training School in Observational Astronomy (ATSOA),” March 2018.
  - Département d’Astrophysique, Géophysique et Océanographie, Université de Liège, Belgium, December 2017 (invited).
  - IUCAA, India, “Thirty Meter Telescope (TMT) Conference,” January 2017.
  - ARIES, India, “ARIES Training School in Observational Astronomy (ATSOA),” March 2017.
  - ARIES, India, “Belgo-Indian Network for Astronomy and Astrophysics (BINA),” November 2016 (poster/flash talk).
  - ARIES, India, “ARIES Training School in Observational Astronomy (ATSOA),” February 2016.
  - ARIES, India, “Tuesday Seminar Series,” May 2016; February 2017.
  - IUCAA, India, “Cloudy Workshop,” September 2015.

WORKSHOPS AND SCHOOLS

- AstroSat Data Analysis Workshop, August 8–11, 2017, ARIES, Nainital, India.
- TMT Workshop on Large Telescope Data Handling, Jan 15–27, 2017, IUCAA, Pune, India.
- Extragalactic Relativistic Jets: Cause and Effect; FERMI Satellite Data Reduction School, ICTS Bangalore, October 14–21, 2015.
- Cloudy Workshop, September 21–26, 2015, IUCAA, Pune, India.
- Workshop on Radio Data Reduction, Radio Astronomy School 2015 (RAS), August 31, 2015, NCRA, Pune, India.

LIST OF PUBLICATIONS

First-Author Publications

1. **Mishra, Sapna**; Khaire, Vikram; Pallikara, Romeo; Narayanan, Anand; Fox, Andrew (minor revision, ApJ) *“Discovery of Weak O VI Absorption in Underdense Regions of the Low-Redshift Intergalactic Medium.”*
2. **Mishra, Sapna**; Fox, Andrew; Smoker, J.; Lucchini, Scott; D’Onghia, Elena; 2025, ApJ, 984, 104. *“The Distance to the Magellanic Stream: Constraints from Optical Absorption along Stellar Sightlines.”*
3. **Mishra, Sapna**; Fox, Andrew; Krishnarao, Dhanesh; Lucchini, Scott; D’Onghia, Elena; Cashman, Frances; Barger, Kathleen; Lehner, Nicolas; Tumlinson, Jason; 2024, ApJL, 976, L28. *“The Truncated Circumgalactic Medium of the Large Magellanic Cloud.”*
4. **Mishra, Sapna**; Muzahid, Sowgat; Dutta, Sayak; Srianand, Raghunathan; Charlton, Jane; 2024, MNRAS, 527, 3858. *“Characterizing Cool, Neutral Gas and Ionized Metals in the Outskirts of Low-z Galaxy Clusters.”*

5. **Mishra, Sapna**; Muzahid, Sowgat; 2022, ApJ, 933, 229. *"Discovery of a Cool, Metal-rich Gas Reservoir in the Outskirts of $z \approx 0.5$ Clusters."*
6. **Mishra, Sapna**; Gopal-Krishna; Chand, H.; Chand, K.; Kumar, A.; Negi, V.; 2021, MNRASL, 507, L46. *"A Search for Blazar Activity in Broad-Absorption-Line Quasars."*
7. **Mishra, Sapna**; Vivek, M.; Chand, H.; Joshi, R.; 2021, MNRAS, 504, 3187. *"Appearance versus Disappearance of Broad Absorption Line Troughs in Quasars."*
8. **Mishra, Sapna**; Krishna, G.; Chand, H.; Chand, K.; Ojha, V.; 2019, MNRASL, 489, L42. *"Are There Broad Absorption Line Blazars?"*
9. **Mishra, Sapna**; Chand, H.; Krishna, G.; Joshi, R.; Shchekinov, Y. A.; Fatkhullin, T. A.; 2018, MNRAS, 473, 5154. *"On the Incidence of Mg II Absorbers along Blazar Sightlines."*

Second-Author Publications

1. † Zhibin, You; **Mishra, Sapna**; Fox, Andrew (to be submitted to ApJ). *"Probing the Small-Scale Routly–Spitzer Effect in the Milky Way Halo."*
2. ‡ Fox, Andrew; **Mishra, Sapna**; Cashman, Frances; French, David; Richter, Philipp; Bordoloi, Rongmon; Lehner, Nicolas; Tumlinson, Jason; Borthakur, S.; 2026, ApJ, 999, 107. *"Low-Metallicity Gas on the Outskirts of the Local Group: The Circumgalactic Medium of Sextans B."*
3. † Kumar, Ritish; **Mishra, Sapna**; Chand, Hum; 2025, MNRAS, 542, 119. *"On the Incidence of Weak and Strong Mg II Absorbers Toward Flat and Steep Spectrum Radio Quasars."*

Co-Author Publications

1. Negi, Vibhore ; Dukiya, Naveen ; Singh Mehra, Gokul ; Ailawadhi, Bhavya ; Dubey, Monalisa ; Filali, Sara ; Hickson, Paul ; Kumar, Brajesh ; Kumari, Priyanshi ; **Mishra, Sapna**; et al. (accepted in PASP). *"Astrometric Calibration of the 4-m International Liquid Mirror Telescope Observations "*
2. Lucchini, Scott; Han, Jess; **Mishra, Sapna**; Fox, Andrew; 2026, ApJ, 1002, 14. *"The LMC Corona Favors a First Passage "*
3. ‡ Dutta, Sayak; Muzahid, Sowgat; Schaye, Joop; **Mishra, Sapna**; Chen, Hsiao-Wen; Johnson, Sean; Wisotzki, Lutz; Cantalupo, Sebastiano; 2024, MNRAS, 528, 3745. *"MUSEQuBES: Mapping the Distribution of Neutral Hydrogen around Low-Redshift Galaxies."*
4. Gopal-Krishna; Chand, K.; Chand, H.; Negi, V.; **Mishra, Sapna**; Britzen, S.; Bisht, S.; 2023, MNRASL, 518, L13. *"Intranight Optical Variability of Low-mass Active Galactic Nuclei: A Pointer to Blazar-like Activity."*
5. ‡ Kumar, B.; Negi, V.; Ailawadhi, B.; **Mishra, Sapna**; Pradhan, B.; Misra, K.; Hickson, P.; Surdej, J.; 2022, JApA, 43, 10. *"Upcoming 4m ILMT Facility and Data Reduction Pipeline Testing."*
6. ‡ Chand, K.; Gopal-Krishna; Omar, A.; Chand, H.; **Mishra, Sapna**; Bisht, S.; Britzen, S.; 2022, MNRASL, 511, L13. *"Intranight Variability of Ultraviolet Emission from Powerful Blazars."*
7. ‡ Ojha, V.; Chand, H.; Gopal-Krishna; **Mishra, Sapna**; Chand, K.; 2020, MNRAS, 493, 3642. *"Comparative Intranight Optical Variability of X-ray and γ-ray Detected Narrow-Line Seyfert 1 Galaxies."*

†: Primary mentor; ‡: Significant contribution in analysis/observations.

CONFERENCE PROCEEDINGS (refereed)

5. Magrini, Laura; with HRMOS Team incl. **Mishra, Sapna** et al. 2026, White Paper for the proposed HRMOS instrument. *"HRMOS: A High-Resolution Multi-Object Spectrograph for the VLT"*

4. Fox, Andrew; Kriss, Jerry; Richter, Philipp; Shull, J. Michael; Cashman, Frances; **Mishra, Sapna**; et al. 2026, White Paper responding to the STScI request for community input. *"High-S/N Quasar Observations with HST/COS: Deep Fields for Spectroscopy"*

3. Vivek, M.; Nair, Akhil; **Mishra, Sapna**; Proceedings of the IAU Symposium No. 378, 2024. *"AGN Outflows and Their Variability."*

2. **Sapna Mishra**; Chand, H.; et al. 2018, Bulletin de la Société Royale des Sciences de Liège, 87, 325. *"Revisiting the Incidence of Mg II Absorbers along Blazar Sightlines."*

1. Chand, Hum; Rakshit, Suvendu; Jalan, Priyanka; Ojha, Vineet; Srianand, Raghunathan; Vivek, Mariappan; **Sapna Mishra**; et al. 2018, Bulletin de la Société Royale des Sciences de Liège, 87, 291. *"Probing the Central Engine and Environment of AGN Using ARIES 1.3-m and 3.6-m Telescopes."*